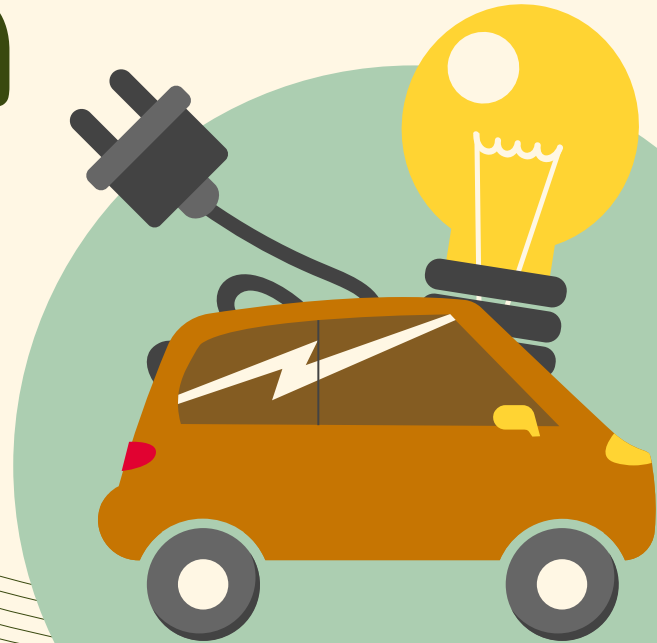


Energy Exploration

Part 1: Energy Transfer in Collision

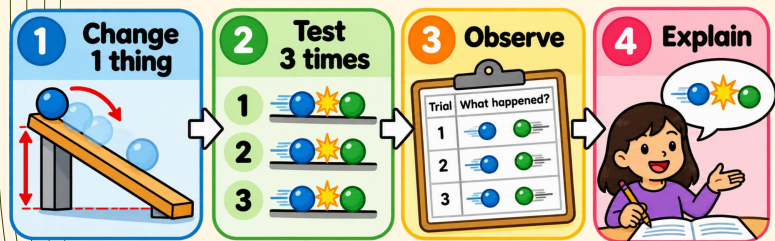
Developed by Yuan Li with AI-assisted drafting. Content, instructional supports, visuals, and design were reviewed and revised by the teacher.



Learning Objectives

Content Objective

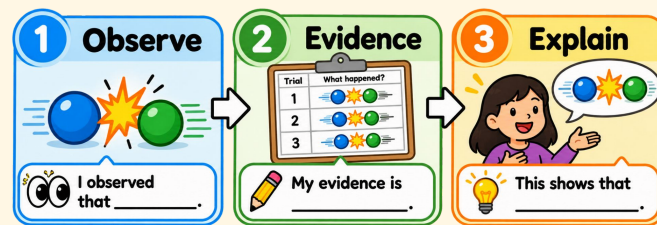
I can investigate what happens when marbles collide and use my observations to explain what I learned.



Pic 1: AI generated

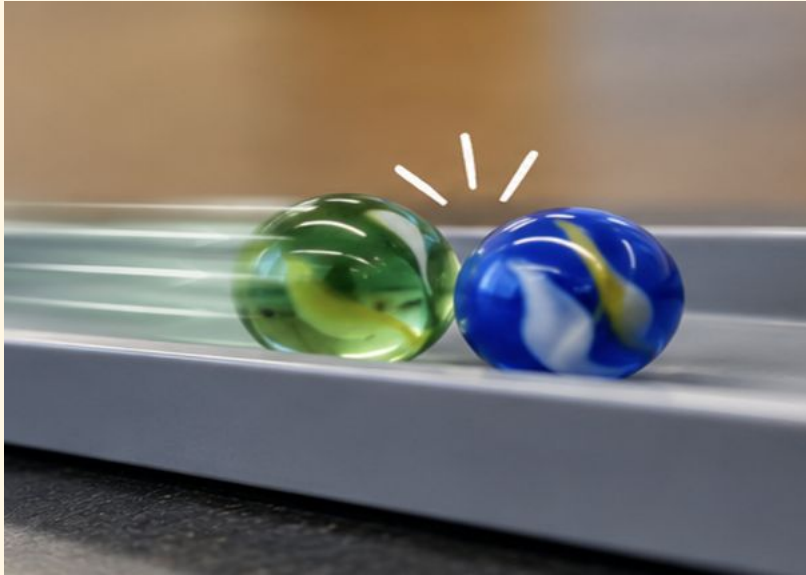
Language Objective

I can use evidence from my observations to explain what happened when the marbles collided.



Pic 2: AI generated

What happens when two marbles hit each other?



NOTICE

What do you see?

WONDER

What do you think will happen after the marbles hit?

Turn & Talk

"I notice_____" "I wonder_____."

Words Scientists Use

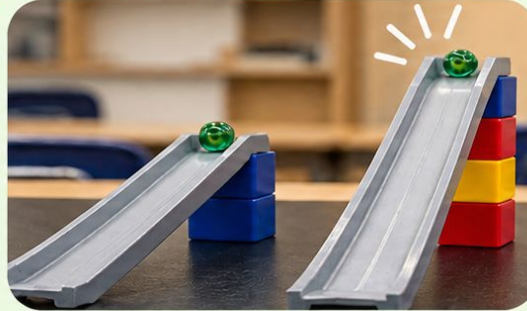


collision

Two objects hit each other.

?

What happened
when they met?

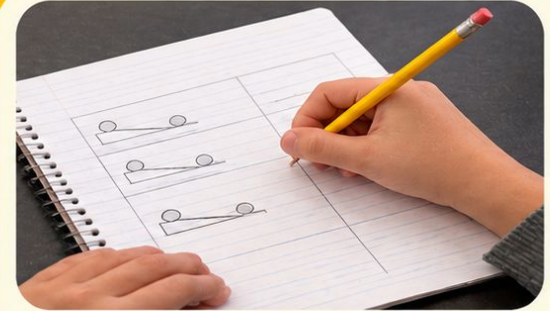


variable

ONE thing we change
in a test.

?

What **ONE** thing
changed?



observation

What you actually
SEE or **HEAR**.

?

What did you
see or hear?

Words Scientists Use

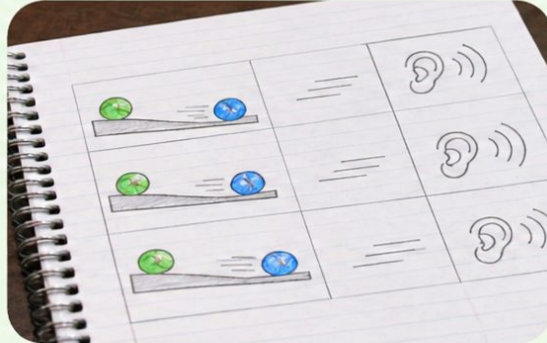


trend

A **pattern** across your trials.



What happened again and again?



evidence

Information from your investigation.



Point to a trial, sketch, or measurement.



energy transfer

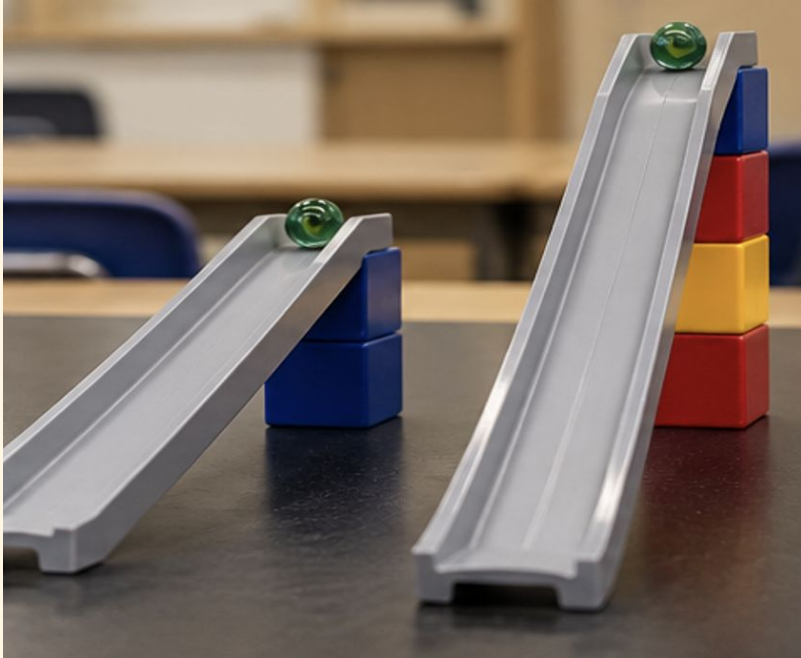
Energy moves from one object or place to another.



What changed after the collision?

Fair Test: Change ONE Variable

If we change many things, we cannot tell what caused the result.



Choose ONE to test:

- Marble size
- Track height
- Release point

Keep the other things the SAME.

Partner frame:

“We will change _____ and keep _____ the same.”

Before we test: Plan & Predict

A prediction is what you think may happen—not an observation.

1. Plan

Step 1: Choose ONE variable

Step 2: Keep other factors the same

Step 3: Predict what will happen

Step 4: Complete at least 3 trials

Step 5: Look for a trend.

2. Predict Frames

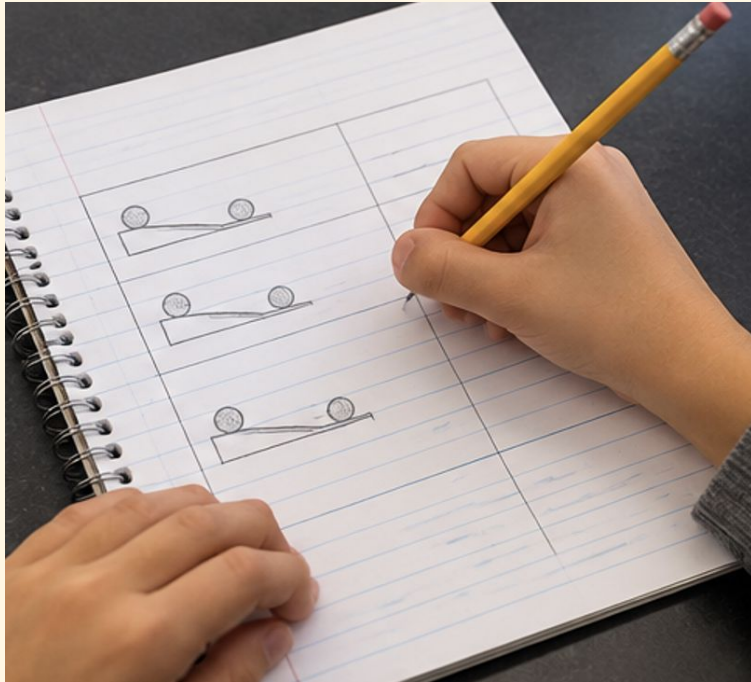
I think that when _____ ,

_____ will _____ .

I also can sketch my idea.

During the test: Observe

Write what actually happens—not what you expected.



Use Your Sense



What did you SEE?



What did you HEAR?

Observation frame:

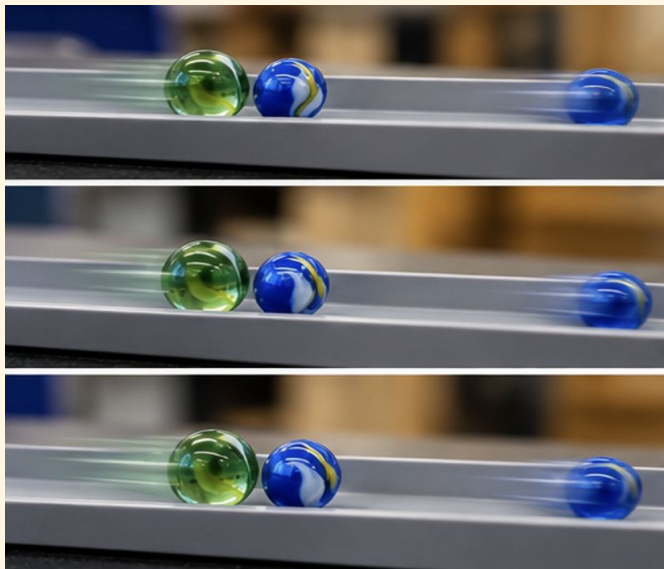
"I observed that _____."

Sketch: Before—Collision—After

Do not change your observation to match your prediction.

After 3 trials: Trend

Look across your records. What pattern do you notice?



1. Find a Trend
“Across our trials,
we noticed _____.”

2. Point to Evidence
“Our evidence is _____.”

Explain Like a Scientist

Claim → Evidence → Reasoning

Claim

What do the results show?



When objects collide,

_____.

Evidence

What did you observe?



In our trials, I observed

_____.

Reasoning

Why does this support your claim?



This supports my claim because

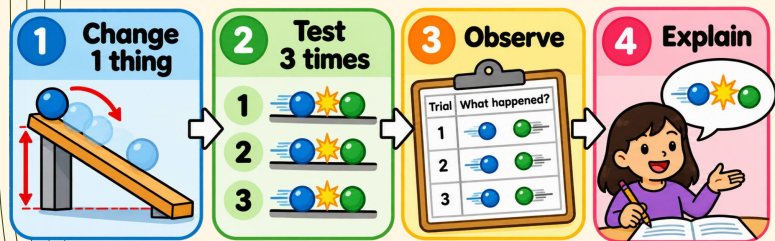
_____.

Before writing: ✎ sketch → 💡 rehearse with a partner → 📝 explain

Learning Objectives

Content Objective

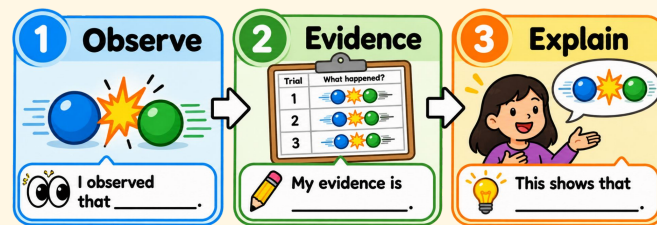
I can investigate what happens when marbles collide and use my observations to explain what I learned.



Pic 1: AI generated

Language Objective

I can use evidence from my observations to explain what happened when the marbles collided.



Pic 2: AI generated

Draw a line from each science word to its meaning.

1



collision

2



variable

3



observation

4



trend

5



evidence

6



energy transfer

MATCH



A

What you actually
SEE or **HEAR**.

B

Energy moves from one
object or place to another.

C

A pattern across
your trials.

D

ONE thing we change
in a test.

E

Information from
your investigation.

F

Two objects hit
each other.

 **Partner Challenge:** Pick one word and explain: “____ means ____.”

Exit Ticket

Review: Can you explain your thinking?

- 1 What variable did your group change?**
- 2 What did you observe across your trials?**
- 3 What trend did you notice?**
- 4 What is your claim? What evidence supports it?**
- 5 Why does that evidence support your claim?**